



Phasor diagram

Phasor diagram shows the magnitude of each phasor quantity in the complex number plane.

Phase angles are measured counter clockwise from positive real axis.

Phasor diagrams give additional insight into the steady state operation of an ac circuit.

Ex. Find V_s using phasor diagram.

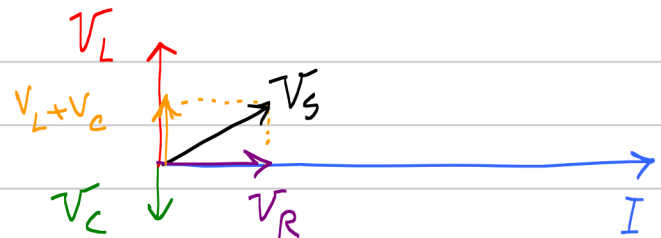
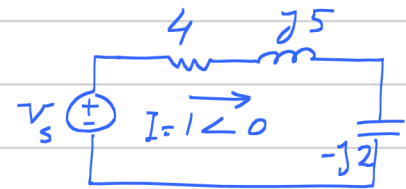
1. start with $I = 1 \angle 0^\circ$.

2. V_R is in phase with I .

3. V_L leads I by 90° .

4. V_C lags I by 90° .

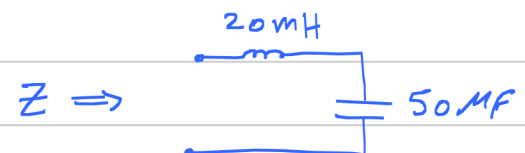
$$5. V_s = V_R + (V_L + V_C)$$



Ex. At what frequency is the series combination of a the inductor and capacitor shown below equivalent to an open circuit? At what frequency it is equivalent to a short circuit?

$$Z = j\omega L + \frac{-j}{\omega C}$$

$$Z = \frac{j\omega^2 L C - j1}{\omega C} = \infty \text{ (open)}$$



$$\Rightarrow \omega = 0 \text{ (dc)}$$

$$Z = \frac{j\omega^2 L C - j1}{\omega C} = 0 \text{ (short)}$$

$$\Rightarrow \omega^2 L C = 1 \Rightarrow \omega = \frac{1}{\sqrt{L C}} = \frac{1}{\sqrt{20 \times 10^{-3} \times 50 \times 10^{-6}}} = 1000 \frac{\text{rad}}{\text{sec}}$$

Sinusoidal steady state power calculations

Power is an important quantity in electric systems. Any electric device has a certain power rating within which the device operates safely.

Instantaneous and average power

$$P(t) = v(t) \cdot i(t)$$

$$V(t) = V_m \cos(\omega t + \theta_v) \quad , \quad i(t) = I_m \cos(\omega t + \theta_i)$$

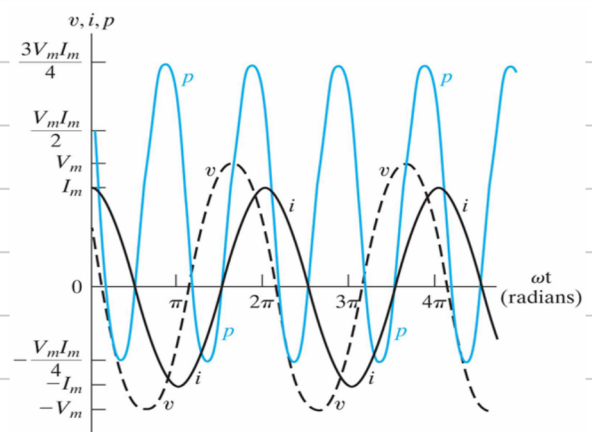
Let's take $i(t)$ as the reference and write:

$$i(t) = I_m \cos \omega t \quad , \quad v(t) = V_m \cos(\omega t + \theta_v - \theta_i) \quad , \quad \text{then}$$

$$P(t) = V_m I_m \cos(\omega t + \theta_v - \theta_i) \cdot \cos \omega t$$

$$\text{using : } \cos A \cdot \cos B = \frac{1}{2} [\cos(A-B) + \cos(A+B)] \quad , \quad \text{then}$$

$$P(t) = \frac{1}{2} V_m I_m \left[\underbrace{\cos(\theta_v - \theta_i)}_{\text{constant}} + \underbrace{\cos(2\omega t + \theta_v - \theta_i)}_{\text{sinusoid at double frequency}} \right]$$



$$P(t) = \frac{1}{2} V_m I_m \left[\cos(\theta_v - \theta_i) + \cos(\underbrace{2\omega t}_A + \underbrace{\theta_v - \theta_i}_B) \right] \quad \text{instantaneous power}$$

using : $\cos(A+B) = \cos A \cos B - \sin A \sin B$,

It can also be written as:

$$P(t) = \underbrace{\frac{V_m I_m}{2} \cos(\theta_v - \theta_i)}_P + \underbrace{\frac{V_m I_m}{2} \cos(\theta_v - \theta_i) \cos 2\omega t}_P - \underbrace{\frac{V_m I_m}{2} \sin(\theta_v - \theta_i) \sin 2\omega t}_Q$$

$$P(t) = P + P \cos 2\omega t - Q \sin 2\omega t \quad (W)$$

$$P = \text{average power} = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) \quad (W)$$

$$Q = \text{reactive power} = \frac{V_m I_m}{2} \sin(\theta_v - \theta_i) \quad (VAR)$$

Power for a purely resistive circuit

For a resistor, the current and voltage are in phase

$$\theta_v = \theta_i \Rightarrow P = \frac{V_m I_m}{2} \cos(\cancel{\theta_v - \theta_i})$$

$$P = \frac{1}{2} V_m I_m$$

For purely resistive ckt

Power for a purely inductive circuit

For an inductor, current lags the voltage by 90° .

$$\theta_v - \theta_i = 90^\circ$$

$$P = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i) = 0$$

$P = 0$

 No average Power

Power for a purely capacitive circuit

For a capacitor current leads the voltage by 90° .

$$\theta_v - \theta_i = -90^\circ$$

$$P = \frac{V_m I_m}{2} \cos(-90) = 0$$

$P = 0$

Power factor

The angle $\theta = \theta_v - \theta_i$ is called power factor angle.

$\cos(\theta_v - \theta_i)$ is called power factor and

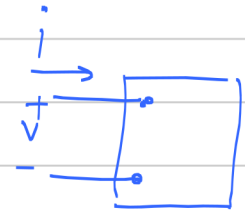
$$pf = \cos(\theta_v - \theta_i)$$

Since $\cos(\theta_v - \theta_i) = \cos[-(\theta_v - \theta_i)]$, knowing the power factor does not indicate if the angle is positive (inductive ckt.) or negative (capacitive ckt.).

We use the terms lagging power factor (current lags the voltage) for inductive circuits, and leading power factor (current leads the voltage) for capacitive circuits.

Ex. Calculate pf, average, reactive and instantaneous powers.

$$V = 100 (\cos \omega t - 45^\circ), \quad i = 20 \cos(\omega t + 15^\circ)$$



$$\theta = \theta_V - \theta_i = -45^\circ - 15^\circ = -60^\circ$$

$$PF = \cos(-60^\circ) = \underline{\frac{1}{2}}, \text{ leading}$$

$$P = \frac{1}{2} V_m I_m \cos \theta = \frac{1}{2} (100)(20) \cos(-60^\circ) = \underline{500 \text{ W}} \quad (\text{abs.})$$

$$Q = \frac{1}{2} V_m I_m \sin \theta = \frac{1}{2} (100)(20) \sin(-60^\circ) = \underline{-866 \text{ VAR}} \quad (\text{del.})$$

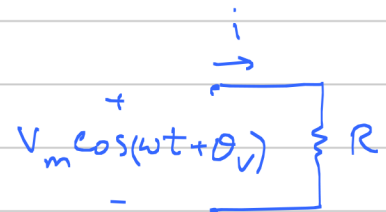
$$P(t) = \underline{P} + \underline{P} \cos 2\omega t - \underline{Q} \sin 2\omega t$$

$$= 500 + 500 \cos 2\omega t + 866 \sin 2\omega t$$

Power calculations in terms of the rms value in a resistor.

$$P = \frac{V_m I_m}{2} = \frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}}$$

$$P = V_{rms} \cdot I_{rms}$$



$$\boxed{P = \frac{V_{rms}^2}{R} = I_{rms}^2 R}$$

average power in a resistor
in terms of the rms value.
(similar to the dc power equation)

Average power in terms of the rms values for any circuit element

$$P = \frac{V_m I_m}{2} \cos(\theta_v - \theta_i)$$
$$= \frac{V_m}{\sqrt{2}} \cdot \frac{I_m}{\sqrt{2}} \cos(\theta_v - \theta_i)$$

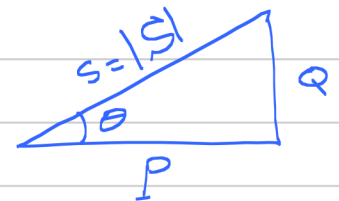
$$P = V_{rms} I_{rms} \cos(\theta_v - \theta_i) = \underline{V_{rms} I_{rms} \cdot pf}$$

Complex power

The complex power is the complex sum of the real and the reactive power.

$$S = P + jQ$$

$$P = \operatorname{Re}\{S\}, \quad Q = \operatorname{Im}\{S\}$$



Power Triangle

P : Real Power (W)

Q : Reactive Power (VAR)

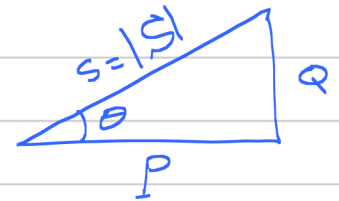
S : Complex power (VA)

$S = |S|$: Apparent Power (VA)

θ : Power factor angle

$$S = |S| = \sqrt{P^2 + Q^2} \quad , \quad \theta = \tan^{-1} \frac{Q}{P}$$

$$\vec{S} = P + jQ = S \angle \theta$$



The real power, P represents the useful power.

The apparent power $S = |S|$, represents the volt-amp capacity required to supply the average power P .

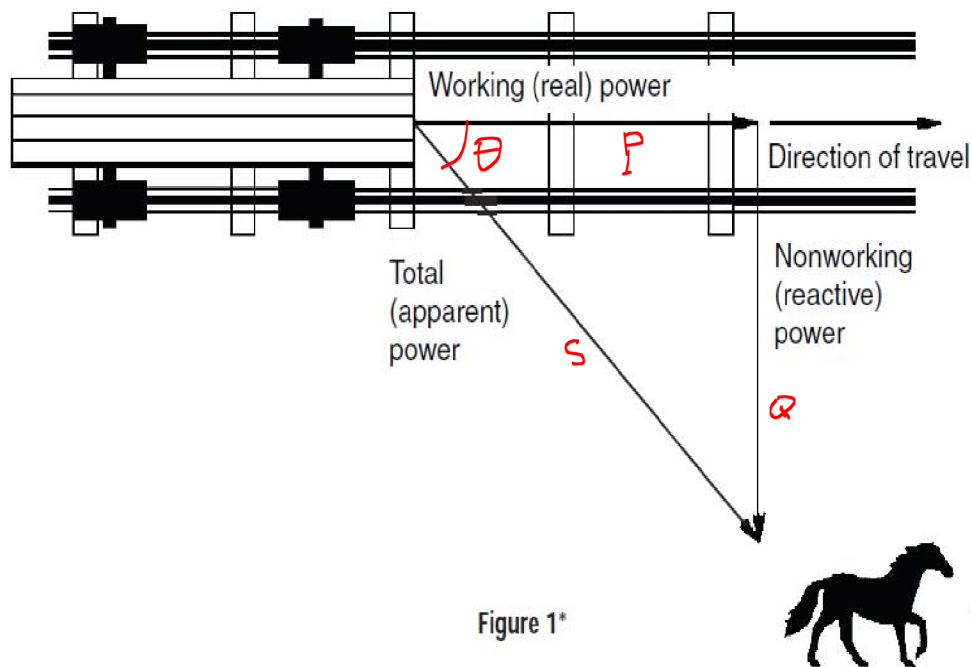
The smaller Power factor angle θ , gives more real (useful) Power from the same consumed Volt-amperes.

It is desirable to have a unity Power factor ($\theta = 0$).

most appliances operate at lagging power factor (inductive).

The power factor of these loads can be corrected

by adding a capacitor across the line feeding the load.



Power triangle for three types of loads

1. Resistive load : $X = 0 \Rightarrow Q = 0 \Rightarrow \theta = 0$

$\Rightarrow \text{pf} = 1$ (unity pf) , no triangle

$$\begin{aligned} P &= S \\ \theta &= 0 \end{aligned}$$

2. Inductive load (RL) : $X > 0 \Rightarrow Q > 0 \Rightarrow \theta > 0$

$\Rightarrow$ Lagging pf



3. Capacitive load (RC) : $X < 0 \Rightarrow Q < 0 \Rightarrow \theta < 0$

$\Rightarrow$ Leading pf

